GLOBAL VIDEO GAME SALES

Exploratory Data Analysis Using Python

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01 | Introduction

Video games are both a creative medium and a global consumer market. This analysis examines how recorded video game sales differ across genres, platforms, publishers and geographic regions. Rather than treating sales rankings as a measure of quality, the study uses them to understand the commercial patterns visible in a historical dataset.

The analysis uses the publicly available Video Game Sales dataset, which contains 16,598 game entries and regional sales estimates in millions of units. I examined the supplied CSV using Python and Pandas, summarised key measures and created charts to communicate the results. The data largely describes an earlier era of physical game sales and should not be interpreted as a picture of the gaming market in 2026.

02 | Dataset and approach

Records	Platforms	Genres	Recorded global sales
16,598	31	12	8,920.44 million

The dataset includes game title, platform, release year, genre, publisher and sales by region. I checked for missing fields and duplicate rows, then aggregated reported global sales across the categories of interest. There are 271 missing release years and 58 missing publishers; these records remain in analyses that do not require those fields. No exact duplicate rows were found.

03 | Key findings

Action is the leading genre. Its reported global sales total 1,751.18 million units, followed by Sports (1,330.93 million). This suggests broad commercial appeal across the titles represented, but the dataset does not establish why customers preferred these genres.

PlayStation 2 leads the platform ranking. PS2 accounts for 1,255.64 million reported units, ahead of Xbox 360 and PlayStation 3. Platform totals reflect the number and popularity of listed titles as well as their sales.

North America contributes the largest regional total. The dataset records 4,392.95 million units for North America compared with 2,434.13 million in Europe and 1,291.02 million in Japan.

04 | Genre and platform performance

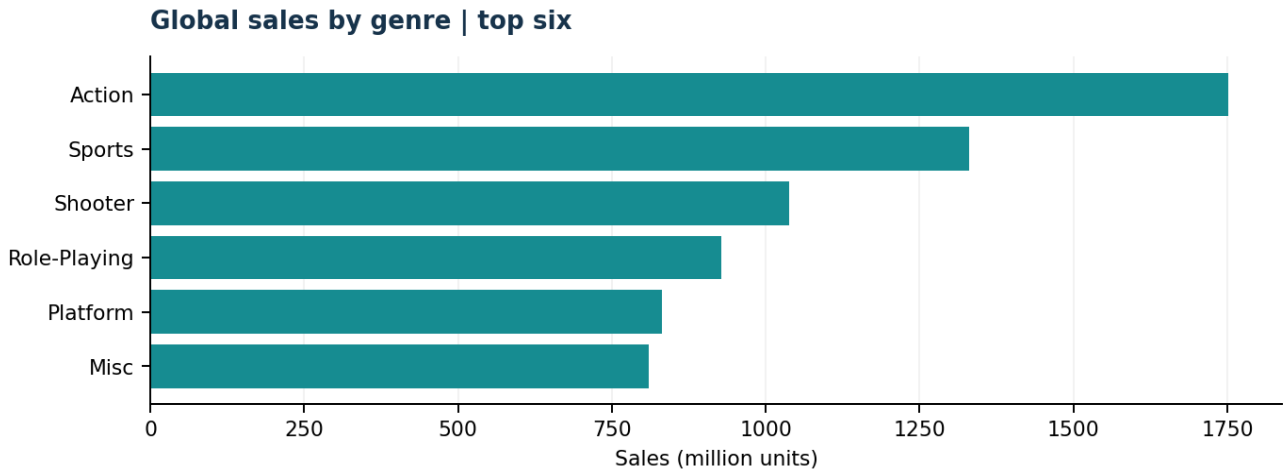


Figure 1. Action and Sports dominate the dataset by cumulative reported sales. A genre total is not the same as average sales per title: larger categories may have more releases.

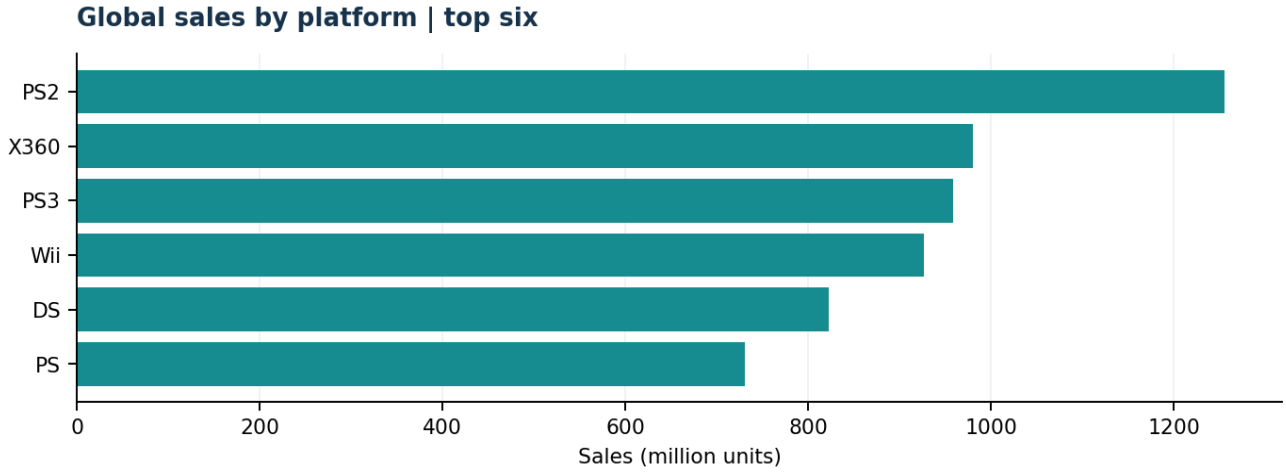


Figure 2. Earlier console generations feature prominently, reinforcing that this is a historical sales dataset rather than a current market tracker.

Interpretation

The leading genres and platforms offer a useful starting point for understanding market concentration. For a fairer comparison, a follow-up analysis could examine median sales per title and the number of releases in each category, rather than relying on cumulative totals alone.

05 | Regional and time trends

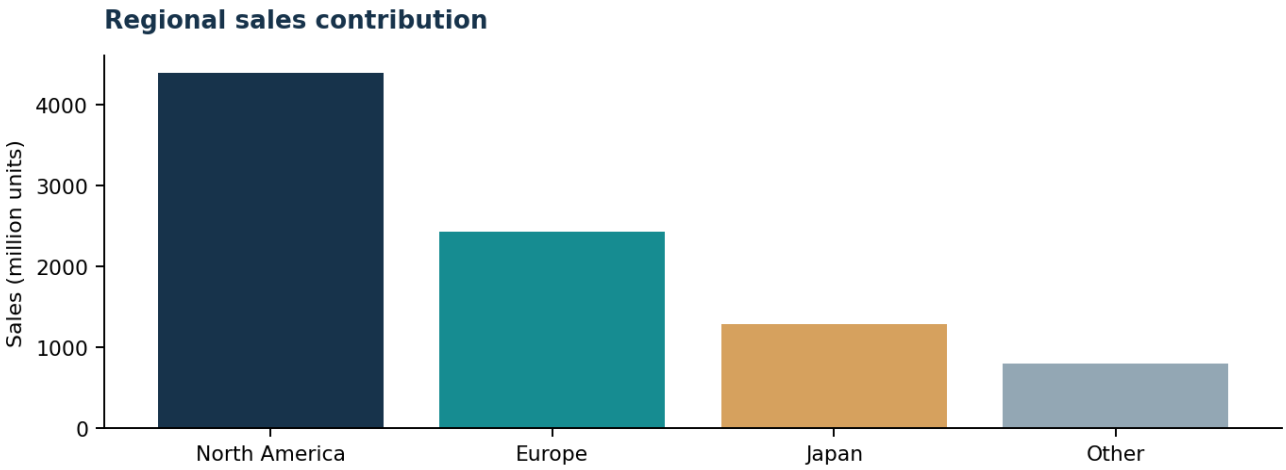


Figure 3. Regional sales estimates are not equal in size. North America accounts for the largest portion of recorded units.

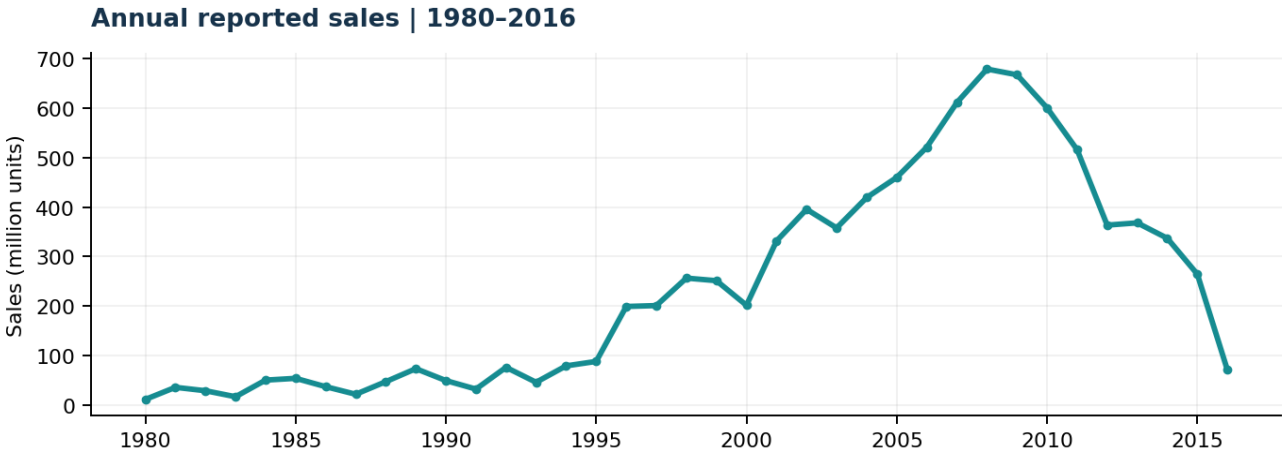


Figure 4. Reported annual sales peak in 2008 at 678.90 million units. Years after 2016 are omitted from this chart because the dataset has sparse, incomplete later-year coverage.

Interpretation

Regional preferences matter when considering where publishers might focus distribution and promotion. The annual curve is best read as a pattern within this particular dataset; it should not be treated as definitive evidence of the entire industry's growth or decline.

06 | Publisher concentration

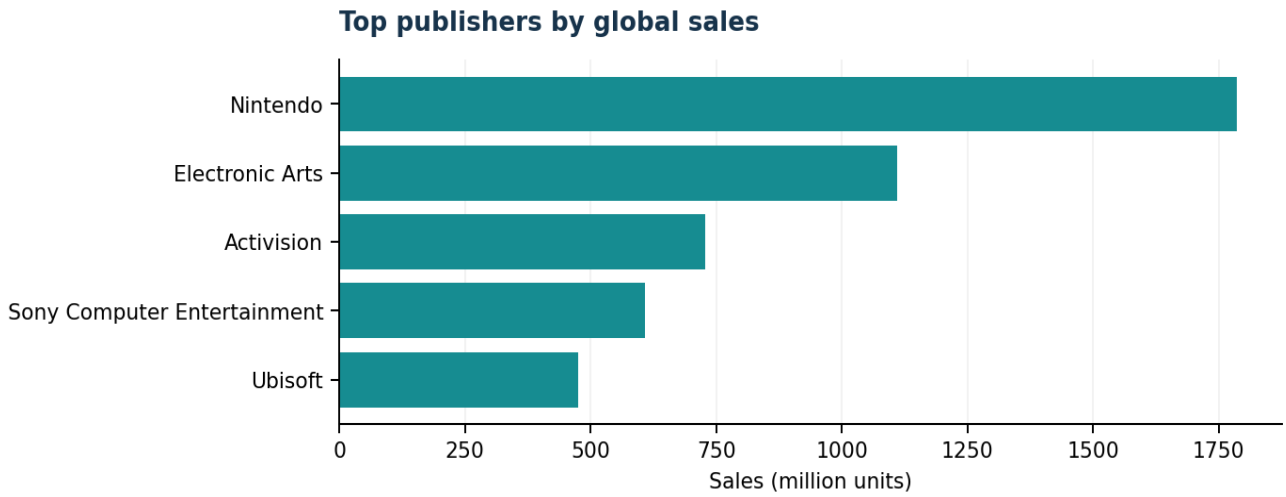


Figure 5. Total reported sales for the five highest-selling publishers represented in the dataset.

07 | Recommendations

Segment analysis by region. Compare the strongest genres within North America, Europe and Japan before drawing conclusions about global consumer demand.

Compare sales per release. Pair cumulative sales with game counts and median title-level sales to distinguish catalogue size from individual title performance.

Use time comparisons carefully. Restrict year-on-year interpretation to periods with adequate coverage and identify missing release dates explicitly.

08 | Limitations and conclusion

This is a secondary dataset of historical game sales estimates. It does not provide revenue, digital downloads, subscriptions or modern live-service activity, and missing years limit time-based comparisons. Regional totals may also differ slightly from global totals because of rounding in the source data.

Overall, the analysis highlights the dominance of Action games, the strong representation of major console platforms and the unequal distribution of recorded sales across regions. Working with the dataset demonstrates the value of careful cleaning, grouped summaries and clear visual communication in turning a large table into an understandable business story.

Data source

Kaggle: Video Game Sales (gregorut). Dataset accessed for this analysis from the supplied CSV.